

Syamantak Kumar

University of Texas at Austin

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University of Texas at Austin

Ph.D. in Computer Science

2023-2027

GPA : 3.97/4.0

University of Texas at Austin

Master of Science in Computer Science

2021-2023

GPA : 3.97/4.0

Indian Institute of Technology Bombay

Bachelors of Technology in Computer Science with Honours

2016-2020

GPA : 9.35/10.0

Research Interests

I am a CS Ph.D. student at UT Austin advised by Prof. Purnamrita Sarkar and Prof. Kevin Tian. My research interests broadly include Statistical Machine Learning, Optimization, Online Learning, Reinforcement Learning.

Publications

1. Dimension-free Score Matching and Time Bootstrapping for Diffusion Models [paper]
Syamantak Kumar, Dheeraj Nagaraj, Purnamrita Sarkar
NeurIPS 2025, San Diego, USA
2. Low-Precision Streaming PCA [paper]
Syamantak Kumar, Shourya Pandey, Purnamrita Sarkar
NeurIPS 2025, San Diego, USA
3. Private Geometric Median in Nearly-Linear Time [paper]
Syamantak Kumar, Daogao Liu, Kevin Tian, Chutong Yang
NeurIPS 2025, San Diego, USA
4. Spike-and-Slab Posterior Sampling in High Dimensions [paper]
Syamantak Kumar, Purnamrita Sarkar, Kevin Tian, Yusong Zhu
COLT 2025, Lyon, France
5. Beyond $\sin^2$ error: Linear-Time Entrywise Uncertainty Quantification for Streaming PCA [paper]
Syamantak Kumar, Shourya Pandey, Purnamrita Sarkar
UAI 2025, Rio de Janeiro, Brazil
6. Oja's Algorithm for Streaming Sparse PCA [paper]
Syamantak Kumar and Purnamrita Sarkar
NeurIPS 2024, Vancouver, Canada
7. Nonparametric Evaluation of Noisy ICA Solutions [paper]
Syamantak Kumar, Purnamrita Sarkar, Peter Bickel, Derek Bean
NeurIPS 2024, Vancouver, Canada
8. Black-Box k -to-1-PCA Reductions: Theory and Applications [paper]
Arun Jambulapati, **Syamantak Kumar**, Jerry Li, Shourya Pandey, Ankit Pensia, Kevin Tian
COLT 2024, Edmonton, Canada
9. Streaming PCA for Markovian Data [paper]
Syamantak Kumar and Purnamrita Sarkar
Spotlight (Top 3%) at NeurIPS 2023, New Orleans, USA
10. From Machine Translation to Code-Switching: Generating High-Quality Code-Switched Text [paper]
Ishan Tarunesh, **Syamantak Kumar** and Preethi Jyothi
Oral Presentation at ACL-IJCNLP 2021, Bangkok, Thailand
11. A comparison of open source libraries ready for 3D reconstruction of wounds [paper]
Syamantak Kumar, Dhruv Jaglan, Nagarajan Ganapathy and Thomas Deserno
Oral Presentation at SPIE Medical Imaging 2019, San Diego, California

Research Experience

Bootstrapped Score Matching

June 2024 - August 2024

Guide: Dheeraj Nagaraj | Research Internship

Google Deepmind

- o Worked as a Graduate Research Intern at Google Deepmind on improving the score matching for Diffusion models by bootstrapping the score function learned at previous timesteps to accelerate learning, reduce bias and improve fairness and overall quality of generation
- o Submitted our work as a research paper at NeurIPS 2025

Feature Extraction for Gaussian Process Regression

July 2019 - July 2020

Guide: Prof. Suyash Awate | Bachelor's Thesis

IIT Bombay

- o Trained end-to-end systems for optimisation of regression parameters using **Deep Kernel Learning**
- o Posed cell-counting (counting cells in a slide image) as a Gaussian Process regression problem and implemented different models for feature extraction, achieving an MAE of 8 on VGG cells dataset
- o Attempted the MICCAI BRATS 2020 challenge for **overall survival(OS)** prediction from MRI scans
- o Trained **U-Net** models with **knowledge distillation** for segmentation, achieving a dice score of 0.92
- o Extracted **radiomic** as well as **CNN-based features** and trained the prediction network with a joint regression plus classification loss to converge to a classification accuracy of 40%

Cross-lingual Embeddings For Code-Switched Text

Jan-July 2020

Guide: Prof. Preethi Jyothi | Research Project

IIT Bombay

- o Experimented with MBERT embeddings for hindi-english code-switched text, pre-training on a multilingual corpus and finetuning on smaller code-switched text to achieve improvements in perplexity
- o Worked on improving training of RNN-based language models on code-switched text, using mixed-review to address the problem of forgetting during finetuning
- o Implemented and developed metrics for evaluation of model-generated code-switched text for perplexity, diversity and text-translation quality

3D Reconstruction of Wounds | Assessment and Diagnosis

May-July 2018

Guide: Prof. Thomas Deserno | Summer Internship

TU Braunschweig, Germany

- o Developed an android application to extract **3D point clouds** by scanning objects using **ARCore**, Google's platform for building augmented reality experiences on smartphones
- o Experimented with Poisson, Alpha Complex, Point Crust, Delaunay Triangulation, Marching Cubes and Convex Hull algorithms for **surface reconstruction** from **sparse point clouds**
- o Compared the results obtained from ARCore with the **Structure from Motion (SfM)** technique to review efficacy of smartphones in generating 3D models for wound assesment

Professional Experience

Microsoft Bing | Core Search and AI

May 2022 - Aug 2022

Applied Scientist Intern

Microsoft, Mountain View

- o Worked on Dense Information Retrieval Models for Bing Search
- o Experimented with early and late fusion architectures to perform Transformer Mixing of Numerical features such as PageRank with Text features for query and document encoders to achieve +0.5 recall
- o Explored query topic annotation features along with numerical features and performed statistical analysis of the model scores to derive key insights into the predictions

Google Maps | Auto-moderation of User Generated Content

August 2020 - August 2021

Software Engineer

Google, Bangalore

- o Designed algorithms to generate signals to detect Hijack & Disguise of businesses on Google Maps
- o Trained lattice regression models for abuse classification based on signals provided by Google Search
- o Performed extensive SQL based analyses to derive insights and statistically verify the algorithms
- o Implemented multiple data processing pipelines in C++ using a MapReduce framework
- o Led discussions with cross-functional teams comprising of 10+ members each, to increase the quality of manual operator labels for abuse classification, with a direct revenue impact

- Built a **generative model** to predict the probability of users making a contribution to Maps
- Modeled user-contributions as a poisson process and performed estimation with **poisson regression**
- **Inferred labels** on unlabelled user edits by aggregating future contributions on each location
- Achieved an **improvement in the coverage of edits** (percentage of edits being labelled) while maintaining similar precision and recall as the original model

Offered a **full-time position** for outstanding contributions during the course of the internship

Coursework

Relevant Courses (UT Austin) : Natural Language Processing, Approximation Algorithms, Statistical Methods, Reinforcement Learning, Advanced Machine Learning, Convex Optimization, Theoretical Statistics, Combinatorics and Graph Theory, Online Learning, Continuous Algorithms, Stochastic Processes, Generative Models in Machine Learning

Relevant Courses (IIT Bombay) : AI and Machine Learning, Automatic Speech Recognition, Fundamentals of Intelligent Learning Agents, Information Retrieval and Web Mining, Digital Image Processing

Teaching and Mentoring Experience

- **Graduate Teaching Assistant** -
 - CS 361S - Network Security and Privacy, with Professor Seth Nielson at UT Austin for Fall 2021
 - EE 461P - Data Science Principles, with Professor Joydeep Ghosh at UT Austin for Spring 2022
- **Undergraduate Teaching Assistant** - Among the **20 students** selected across all batches for teaching a class of **47** first-year students for the undergraduate course on Quantum Physics. Coordinated with the Physics Dept. to conduct regular **tutorial sessions & evaluate exam papers**
- **Teaching Assistant** - Selected as a Teaching Assistant for the Sandbox@Alphabet team's "Linear Algebra for Machine Learning" course, engaging in a chat room of 700+ participants and leading small independent break-out sessions with attendees to discuss problems and exercises.
- **Institute Student Mentor** - Mentor to 12 freshmen students, helping them cope with the curriculum and focusing on their holistic development, selected via a rigorous procedure comprising of interviews
- **Department Academic Mentor** - Mentor to 6 sophomores for their academic & general concerns and helping them cope with the curriculum. Guiding 2 students as part of the Academic Rehabilitation Programme to help improve their academic performance

Professional Service

Served as a reviewer for NeurIPS 2024, 2025, ICML 2025, JMLR 2024, TMLR 2025

Notable Projects

Adversarial Examples for Keyword spotting

July 2019 - November 2019

Guide: Prof. Preethi Jyothi | Course Project

IIT Bombay

- Used **AdvGAN** to generate adversarial examples for keyword spotting by modifying training examples
- Enhanced the performance and robustness of **keyword spotting systems** by training on such examples

Top-k Tournament Ranking From Pairwise Preferences

July 2019 - November 2019

Guide: Prof. Shivaram Kalyanakrishnan | Course Project

IIT Bombay

- Designed and empirically evaluated algorithms for tournament ranking from pairwise preferences
- Explored schemes for fully-sequential sampling by modelling it as a stochastic **multi-armed bandit**
- Theoretically proved correctness under the **Probably Approximately Correct (PAC)** framework

Artistic Style Transfer using Image Processing

July-Nov 2018

Guide : Prof. Suyash Awate | Course Project

IIT Bombay

- Implemented a **transfer technique for re-styling images** based on migrating the style from a given image to the content of another image

- Formulated the problem as an **expectation-maximization** optimisation, with alternating updates
- Used iterated reweighted least squares (IRLS) to approximate the exact solution and improve efficiency
- Achieved results comparable to the style-transfer by CNNs, using only **image processing paradigms**

Toonify - A Cartoonifying Application

July-Nov 2017

Guide : Prof. Kavi Arya | Course Project

IIT Bombay

- Designed an android-based application to perform basic **Image-Processing** operations including cartoonifying an input image, generating a pencil sketch, denoising an image and basic image editing
- Performed **Cel-Shading** for cartoonification using the **Canny-Edge Detection** algorithm for edge detection and **Bilateral Filtering** for smoothing the image with the help of OpenCV4Android
- Implemented an algorithm for generating a pencil sketch of an image using **Color-Dodge Blend**

Chinese Checkers AI

Jan-April 2018

Prof. Amitabha Sanyal | Course Project

IIT Bombay

- Used **minimax strategy** with **alpha-beta pruning** to develop an efficient AI for Chinese Checkers
- Developed a testing interface to conduct **AI vs AI matches** to check the efficacy of our algorithm

Compiler Design

Jan-April 2019

Prof. Supratim Biswas | Course Project

IIT Bombay

- Developed a **lexer** for a subset of C-language using flex (lexical analyser generator)
- Designed a **LALR(1) grammar** for the parser and generated Abstract Syntax Trees using yacc
- Generated three-address intermediate and MIPS assembly codes from **Control Flow Graphs(CFG)**

Awards and Scholastic Achievements

- Secured **All India Rank 30** in **JEE-Main** out of over 1.2 million candidates (2016)
- Achieved **All India Rank 370** in **IIT JEE-Advanced** out of over 140,000 candidates (2016)
- Awarded **AP grade** for excellent performance in **Quantum Physics** and **Physical Chemistry** (2016)
- Received **Gold Medal** for being placed among the **top 38** students in **INChO**, HBCSE (2016)
- Amongst the **top 300** students in **INPhO**, Indian National Physics Olympiad, HBCSE (2015,2016)
- Amongst the **top 300** students in **INAO**, Indian National Astronomy Olympiad, HBCSE (2016)
- Amongst the **top 47** students to attend the OCSC of International Junior Science Olympiad (2013)
- Awarded the prestigious **KVPY Fellowship** by DST, Govt. of India with **All India Rank 17** (2014)
- Awarded **NTSE Fellowship** by NCERT, Govt. of India with **All India Rank 29** (2012)
- Awarded **Amazon AI PhD Fellowship**, selected as one of the 15 students at UT Austin (2025-)

Extracurriculars

- Attended **Vijyoshi camp** conducted by IISER, Kolkata which serves as a forum for interactions between bright young students and leading researchers in fields of science and mathematics
- Part of School **Table-Tennis** Team which **won** the **District-Level Tournament** conducted by the District Sports Organisation, Rajasthan
- Stood **first** in the city finals of **Aqua-Regia - The Science Quiz** certified as the largest quiz at one location by the Guinness Book of World Records